Sound speed and resonance

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use travel time and explain when resonance occurs to interpret sound.

SUCCESS CRITERIA - I can...

1. Account for round-trip distance in an echo.
2. Identify resonance as a response near a natural frequency.
3. Use an appropriate wavelength relationship.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

Two waves meet the same opening with wavelengths **3 cm** and **5 cm**. Which generally diffracts more? **Explain the relevant comparison**.

Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: $d_{echo} = v\,\Delta t/\mathbf{2}$; v = **340 m/s**.
Echo time is a round trip. Resonance occurs near a natural mode under suitable damping.

Worked reasoning: In an ideal tube open at both ends, the fundamental wavelength is twice its length. For L=**0.5 m** and sound speed **340 m/s**, f=**340/1**=**340 Hz**.

Your turn: Sketch the fundamental with displacement antinodes at both ends.

open-open tube

Draw a displacement pattern; label nodes and antinodes.

Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

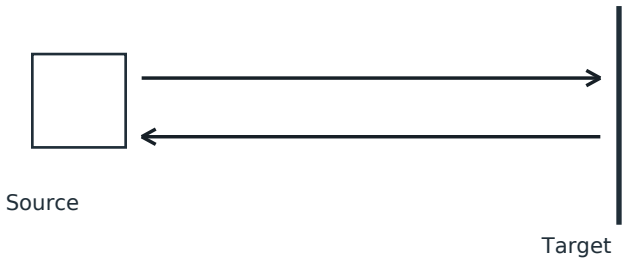
An echo returns from a wall.
1. **Draw the outgoing** and returning path.
2. Why does sound's total travel distance exceed wall range?



Marzano check - circle: ① ② ③

04 Apply the physics

In a scaled echo simulation, a pulse travels at **10 m/s** and returns after **4 s**.
1. **Label both legs of the** trip.
2. **Calculate the total path distance** and the one-way distance to the wall.



Marzano check - circle: ① ② ③

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Continue down each column.

05 Explain your reasoning

Why does a resonant response not mean that any object can respond strongly to every frequency?

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A heat engine receives **600 J** heat and outputs **180 J** work per cycle. **Find efficiency** and rejected heat. **Label energy entering**, leaving and stored.

Draw and label all energy transfers across the boundary.

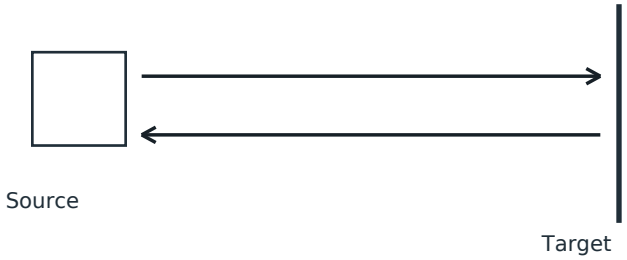
Chosen system

Stored energy change: _____ J

Marzano check - circle: 1 2 3

07 Exit check

An echo’s round-trip path is **60 m**. **How far** away is the wall?



Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

Return to the worked example in problem **2**. **Compare fundamental frequencies** for ideal open-open tubes of lengths **0.5** and **1.0 m** at the same sound speed.

open-open tube

Draw a displacement pattern; label nodes and antinodes.

Marzano check - circle: 1 2 3